Fiscal Year:	FY 2005	**Task Last Updated:**	FY 11/16/2005
PI Name:	Towe, Bruce Ph.D.		
Project Title:	A Microfluidic Bioreporter for Exploratory Probes		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85287-9709	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
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No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	4
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Towe, Bruce		
COI Name (Institution):	Pellis, Neil (NASA - JSC)		
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	An innovative approach to a compact bioreporter chip is being investigated whose function will be to act as a probe for spacecraft, extraplanetary, and earth environments. It will give information that is difficult or impossible to derive from combinations of physical measurements since it gives the added relevance of environmental effects on life. The proposed hardware device will support a small (50 microliter) culture of bacteria in a compact plastic package that would fit into a shirt pocket. Microfluidics controlled by a microprocessor will feed and maintain the bacteria and monitor changes in their metabolic biochemical profiles as an indicator of environmental stressors. The device employs an integrated microchemistry system to monitor culture biochemistry and control feeding based on this. By comparing cellular metabolic profiles in terms of rates and quantity of uptake of nutrients and its production of primary and possibly secondary metabolites, this work will profile cellular responses to environmental stressors. Living bacteria as bioreporters can present a wide variety of sensitivities and unique responses to environmental stresses. Applications might include monitoring radiations, magnetic and geomagnetic fields, response to microgravity, as well as to gaseous,		

Task Description:	aqueous, or liquid-carried filtrates of soil or other aspects of the environment. The objective of this project is to: 1. Employ microfluidic technologies to create perfused cell microcultures on a compact monolithic plastic substrate. 2. Integrate the microculture with a series of microfluidic-based biochemistry sensors to monitor the cellular uptake of nutrients and products of cellular metabolism. 3. Implement intelligent computer control and data capture from the bioreporter. 4. Characterize the biomolecular signatures of the test cell lines by exposure to control as well as challenging environments. This is an interdisciplinary project involving bioengineering and cell science. We are interested in the development and use of bioreporters as both a probe and a tool that would ultimately allow the posing of multiple different scientific hypotheses of importance not only in space environments but earth-based applications as well. Biochemistry monitoring of bacterial metabolism or possibly other types of cells is a key aspect of the envisioned bioreporter system. Microchemistry sensors employed in this work are based on a color-changing or fluorescence-producing chemical reaction that occurs when a dialyzed sample of cell culture effluent is mixed with a specific reagent. The microfluidic hardware configuration of sampling, mixing, and the measurement of color, absorbance, or fluorescence can be adapted to measure the concentration of nearly any low molecular weight biochemical for which there exists a sensitive optical test. Besides glucose, oxygen, carbon dioxide, and pH, the system has the possible ability to measure glutamine, other sugars, urea, uric acid, lactic acid, acetic acid, ethanol, ammonia and other biochemicals. Man's venture into new and little understood environments of space and other planets presents a host of unknowns about how life will react and adapt. Simple physical instruments cannot easily determine the effect of environments on the living organisms because of life's complexity, ability to self-repair, and to change in order to accommodate to the environment. Instrumented probes will be used in space and extraplanetary venues to measure the physical aspects of the environment such as its gravity, atmosphere, radiation environment, and ultimately the suitability of its soil to support life. However we have no good way of predicting the aggregate of effects of multiple conditions of which none individually is lethal. To assess these environments, bioreporter systems will be a key element in the human exploration and development of space. A bioreporter combines living cells such as bacteria or mammalian cells with one or more sensors so as to provide information on how the environment affects cellular processes. Bioreporters can give insight into how the environment's chemical composition, atmosphere, light, and radiation fields interact with the complexity of life processes rather than just a single environing physical condition. When used to monitor and then alarm upon a specific circumstance or hazardous condition, a bioreporter would be called a biosentinel. Both bioreporters and biosentinels are enabling technologies that have the potential to serve an essential role in the initial and intermediate phases of the exploration for planetary missions. Their advantages lie in their direct relevance to human exploration of space, compact size, cost effectiveness, and adaptability to meet specific exploration objectives and goals.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These types of living-cell biosensors have a potentially broad application to environmental monitoring on earth as well as in space. They have the potential to act as sensors for conditions that affect living things from a variety of causes that would be missed by sensors for specific substances, or toxic conditions. Living-cell biosensors also have the potential to respond to the presence or absence of certain pathogenic organisms in the environment and so may useful in bioterrorism detection and early warning.
Task Progress:	The project has substantially completed its goals and the work has produced data supportive of a practical way to conduct space environment molecular biology. The microfluidic system uses bacterial models housed in microfluidic instrumentation accomplished in very small format devices. These devices use micropumps and valves to feed, maintain, and optically monitor growing cells in a device size that might fit in the palm of the hand, exclusive of external laptop computer control. Different realizations of the bioreporter have shown an ability to track the activity of reporter genes by way of their cellular luminescence, RFP fluorescence, optical density, and pH in growing cell cultures of volumes of 50 microliters and less. Integrated fluid control and computer based feedback have been successful in demonstrating the operational lifetime of such systems to week-order intervals with the calculated potential for month-order continuous operational lifetimes. The space bioreporter as developed has shown capability to track bacterial reporter genes that respond to environmental stressors. We have used low concentrations of ethanol, ultraviolet light, and chemical stressors and inducers to demonstrate the responses of such cell-based biosensors to differing environmental conditions that might be suggest their responses to various space conditions. We have demonstrated the unique idea of cycling the bacterial numbers in microcultures by a process of automated growth and then flush-out with the remnant acting as innoculum for the next culture cycle. This approach prevents overgrowth of bacteria in microculture but more importantly keeps the bacteria in a continual log phase of growth where they would be most sensitive to environmental stress.
Bibliography Type:	Description: (Last Updated: 06/19/2005)
Presentation	Towe, B.C., Martineau, R., Paccini, M., Stout, V.,Pellis,N. "A Microfluidic System for Space Exploratory Probes" NASA IWG Meeting, February 2005 Feb-2005